

First name:

Last name:

Student ID:

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Problem 1. *Solve the heat equation*

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

for a very long bar ($-\infty < x < \infty$) with initial conditions:

$$u(x, 0) = f(x)$$

using Fourier transforms and the method of convolution to show that

$$u(x, t) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^{\infty} f(p) \exp \left\{ -\frac{(x-p)^2}{4c^2 t} \right\} dp.$$

[10 marks]

Hints: You may use the following results directly in your solution. These results have been provided to you roughly in the same order that they should appear in the solution, so use this list to help guide you through the steps of the solution as well.

- If $\hat{u} = \mathcal{F}(u) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u(x, t) e^{-i\omega x} dx$, then

$$\mathcal{F}(u_t) = \frac{\partial \hat{u}}{\partial t}, \quad \text{and} \quad \mathcal{F}(u_{xx}) = -\omega^2 \hat{u}.$$

- Solution to a D.E. of the form $\frac{dh}{dt} = kh$ looks like $h(t) = Ce^{kt}$. You will have to make the appropriate adjustments for the problem at hand.
- The initial condition yields the relationship $\hat{u}(\omega, 0) = \hat{f}(\omega)$, where $\hat{f} = \mathcal{F}(f)$.
- If $\hat{g}(\omega) = \frac{1}{\sqrt{2\pi}} e^{-c^2 \omega^2 t}$, and $g = \mathcal{F}^{-1}(\hat{g})$, then

$$g(x) = \frac{1}{2c\sqrt{\pi t}} e^{-\frac{x^2}{4c^2 t}}.$$

- If $\hat{f} = \mathcal{F}(f)$ and $\hat{g} = \mathcal{F}(g)$, then

$$\int_{-\infty}^{\infty} \hat{f} \cdot \hat{g} \cdot e^{i\omega x} d\omega = (f * g)(x) = \int_{-\infty}^{\infty} f(p) g(x-p) dp.$$

Solution.

(Solution continued)

□

END OF QUIZ 9